Deep Learning Models for estimating performance loss in PEM Fuel Cells under CO contamination

Candidate

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FOR A HUMAN-CENTERED AI

CLEANER
Clean Heat and Power from Hydrogen

Objective

The objective of this project is to assess whether AI models can accurately predict the voltage evolution of PEM fuel cells under carbon monoxide contamination, with the aim of supporting prognostics and health management applications.

PEM fuel cells

Proton Exchange Membrane fuel cells

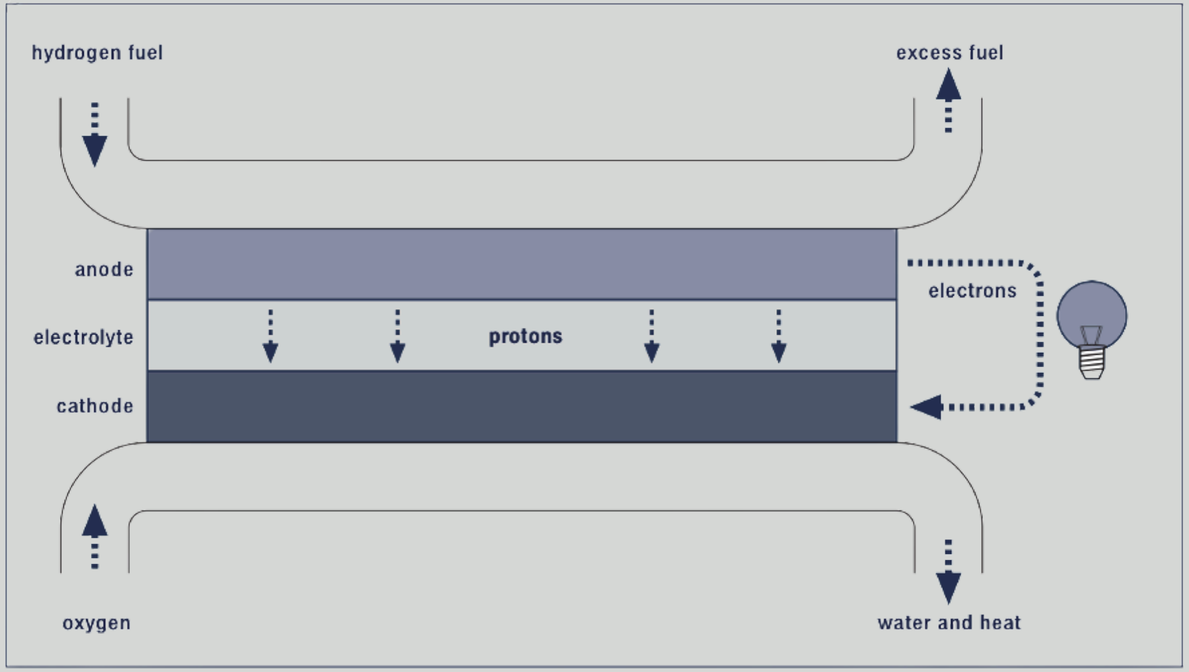
- Generate electricity through the reaction between hydrogen and oxygen, producing heat and water as by-production.
- Widely considered promising for transport and distributed energy applications.

PEM cells are sensitive to hydrogen purity

- Hydrogen purity depends on the production methods and may contain trace of other gases such as Carbon Monoxide (CO).

Why carbon monoxide is critical

- CO binds strongly to the anode surface of PEM fuel cells.
- This attachment blocks the sites where hydrogen should react, causing electricity loss.
- This phenomenon is known as CO poisoning.



Dataset

Structure of the experiments

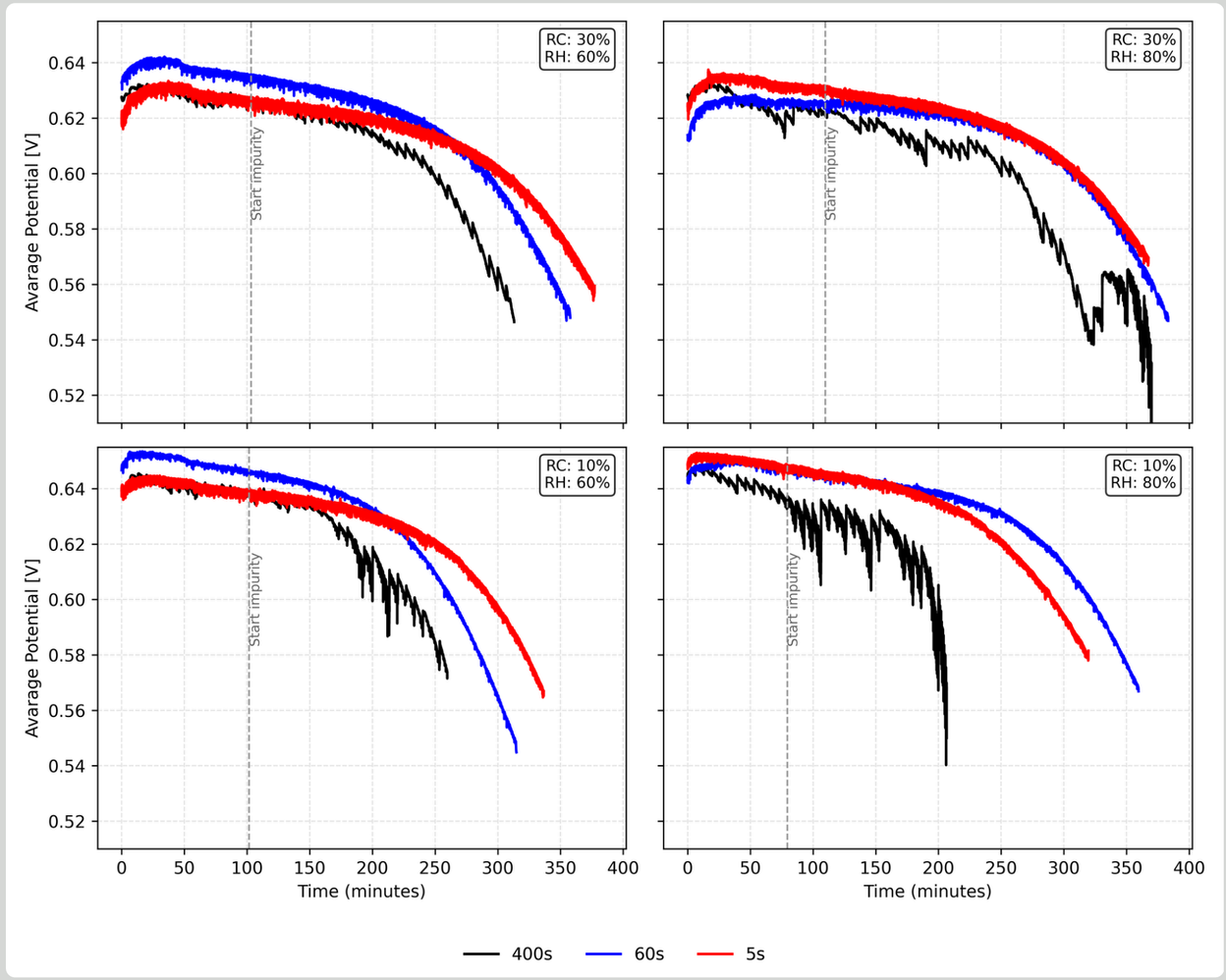
- 12 experiments recorded on a 10-cell PEM fuel cell stack under stable operating conditions.
- Stabilization phase with pure hydrogen followed by controlled CO exposure (represented by the dotted line).
- measurements recorded every second.
- Target variable: average stack voltage (Y).
- Feature set: more than 100 operating parameters (X).
  Ex: pressure, gas composition, humidity, electrical load.

Experimental settings

- Purge Time (5s, 60s, **400**s): How often the system performs a brief cleaning step using fresh hydrogen. Longer intervals allow more impurities to accumulate and lead to faster degradation.

- Recirculation Rate (10%, 30%): Indicates how much of the hydrogen is reused instead of being discharged.

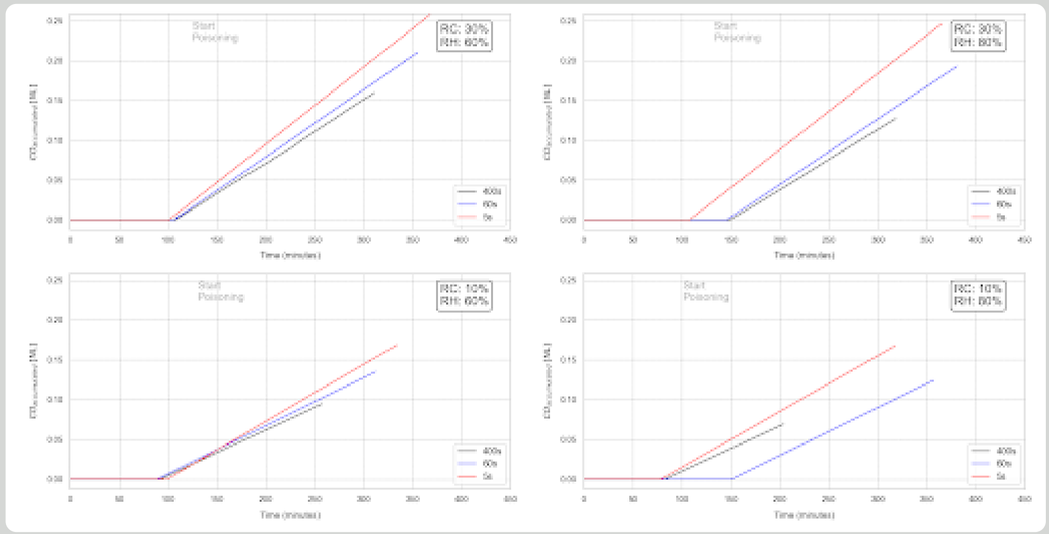
- Relative Humidity (60%, 80%): Indicates how much humidity is present during operation.



Two keys parameters

CO accumulated – *physical approach*

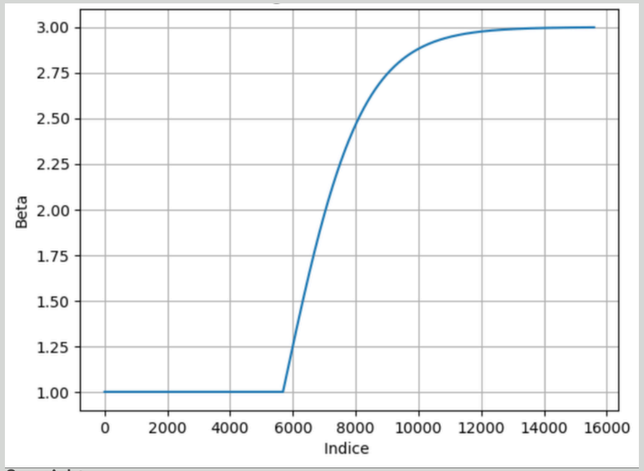
- It approximates the amount of CO that accumulates on the anode over time.
- It is a simplified approximation: in real-world conditions, the CO entering the cell is unknown.



CO Accumulated

$ß_t$ weight – *numerical approach*

- β is a custom weighting parameter added to the (cost) loss function during training, guiding how prediction errors are penalized.
- It increases the importance of the degradation phase after CO injection, encouraging the model to focus on the voltage drop rather than on the initial stable operating period.
- Before CO injection ß = 1.
- After CO injection it increases following a tangent-shaped curve, reaching a maximum value of 3.

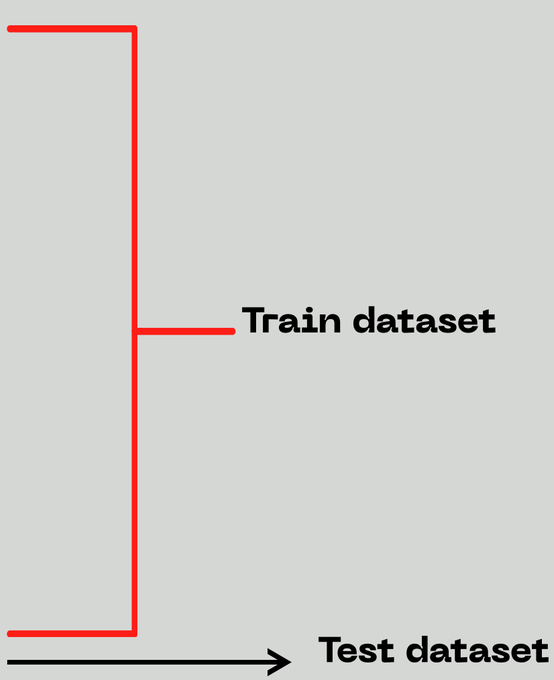


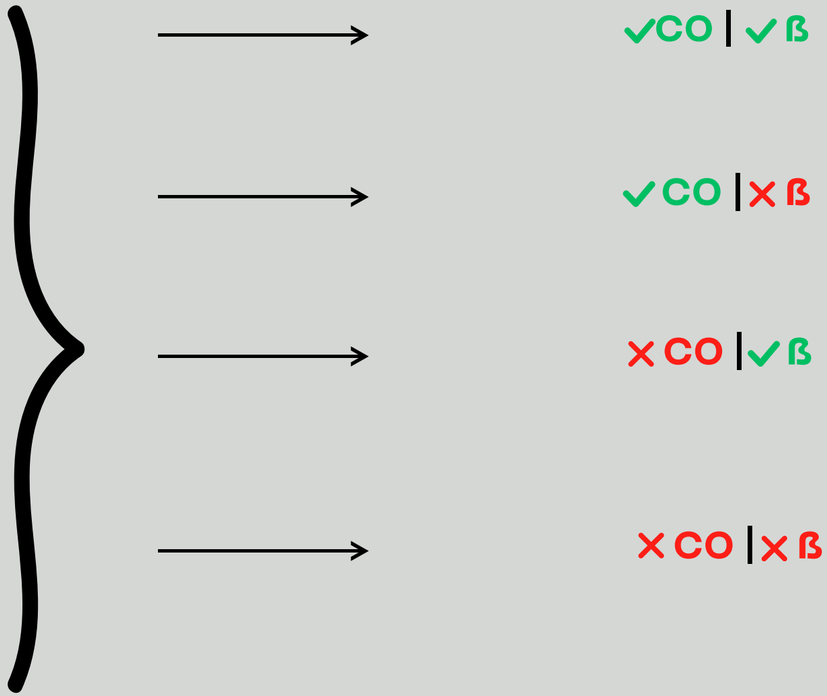
$ß_t$ weight

Configurations

48 dataset configurations were evaluated by training the models on 11 experiments and testing them on the remaining one, repeated across all 12 experiments and all 4 input settings (CO accumulated and ß$_t$ included or excluded)

EXPERIMENT NAME
5S 10RC 60RH
5S 10RC 80RH
5S 30RC 60RH
5S 10RC 80RH
60S 10RC 60RH
60S 10RC 80RH
60S 30RC 60RH
60S 10RC 80RH
400S 10RC 60RH
400S 10RC 80RH
400S 30RC 60RH
400S 10RC 80RH





Models

Data Preprocessing

- Feature selection based on physical relevance
- Removal of highly correlated variables
- Standardization of all features

Metrics:
MSE
RMSE
MAE
R^2

Polynomial Regression

Static regression model used as a benchmark to compare against sequential architectures.

Outcomes:

- Captures the general downward trend but cannot reproduce the spikes introduced by the purge-time parameter.
- Performance varies widely across configurations.
- Consistently shows the lowest predictive accuracy among all models.

LSTM — GRU

Both models are designed to handle **temporal** sequences
They learn the evolution of the voltage signal over time, capturing trends and transitions
Able to retain relevant past information and adapt to new conditions at each timestep.

Outcomes:

- Learns the overall degradation trajectory, clearly outperforming the polynomial baseline.
- Shows delayed or smoothed reactions to purge-time–induced spikes.
- Accuracy depends on operating conditions: it is less consistent than GRU in irregular experiments.

Outcomes:

- Provides the **most accurate** and stable predictions across operating parameters.
- Reproduces both the initial steady regions and the non linear degradation after CO injection with high fidelity.
- More robust than LSTM under irregular regimes (e.g., 400s purge-time cycles), maintaining coherent dynamics without over-smoothing.

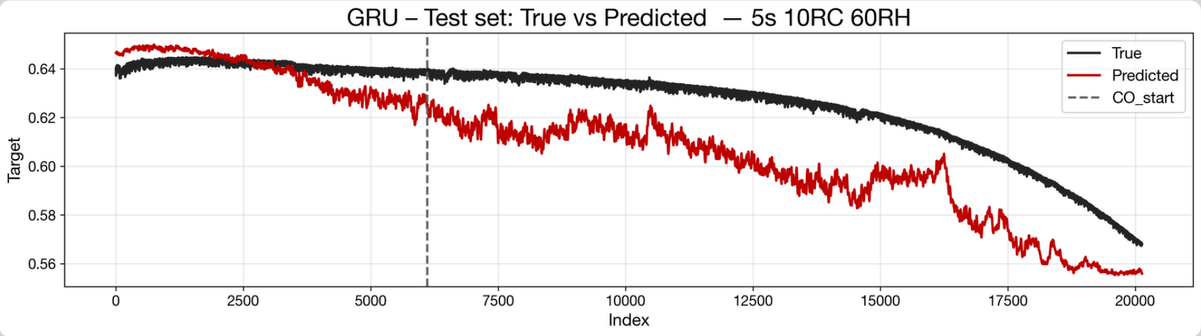
To obtain these results, 144 configurations were evaluated: 48 dataset setup x 3 different models.

Leave one experiment out

Test set: 5s 10RC 60RH

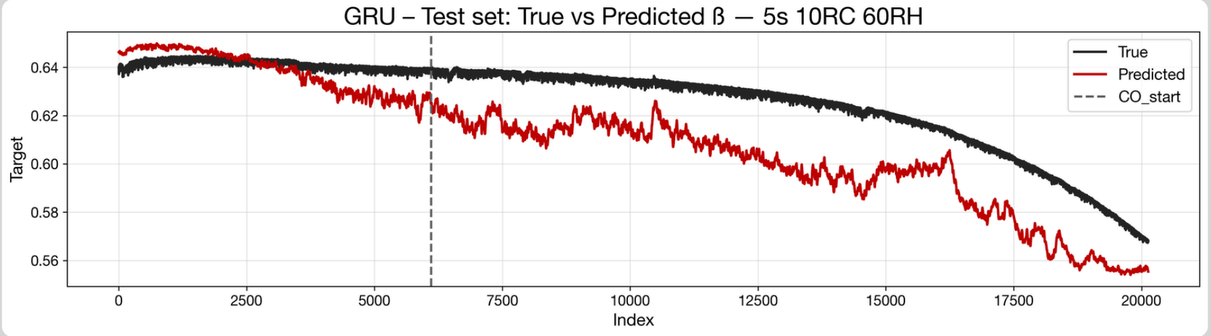
✗ CO | ✗ ß

MSE: 0.881; MAE: 0.830; RMSE: 0.9386; $R^{2:}$ -0.3568



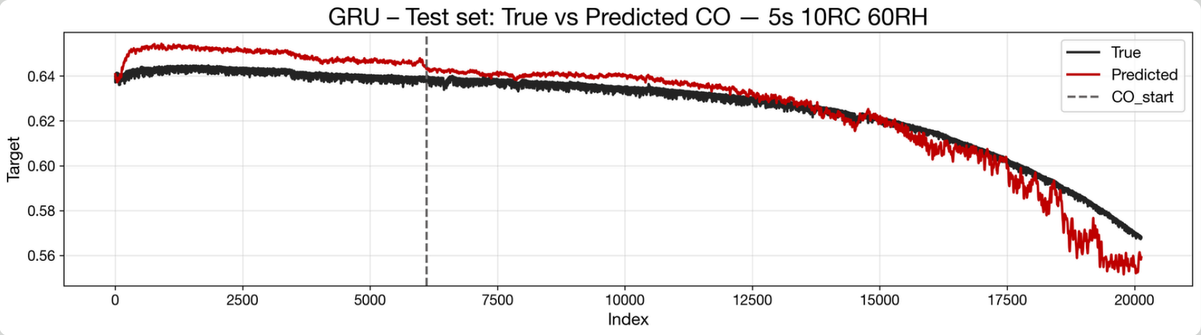
✗ CO | ✓ ß

MSE: 0.7886; MAE: 0.7911; RMSE: 0.8802; $R^{2:}$ -0.2144



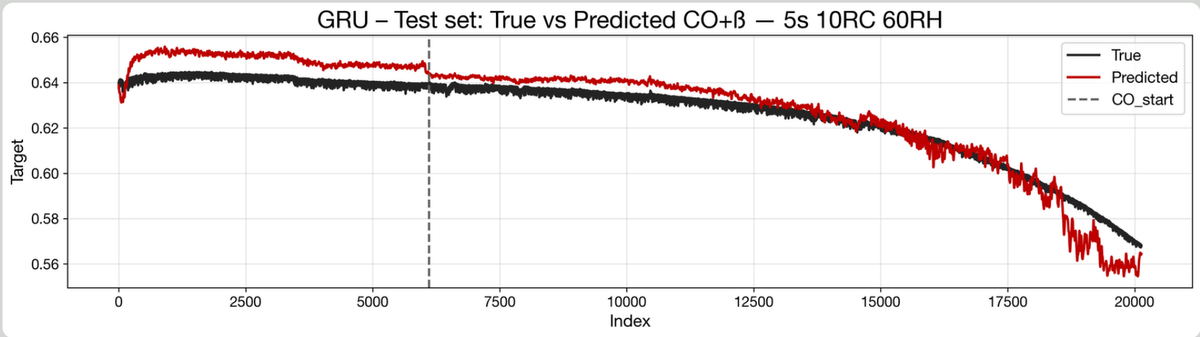
✓ CO | ✗ ß

MSE: 0.1077; MAE: 0.2646; RMSE: 0.3282; $R^{2:}$ 0.8344



✓ CO | ✓ ß

MSE: 0.1001; MAE: 0.2633; RMSE: 0.3163; $R^{2:}$ 0.8459

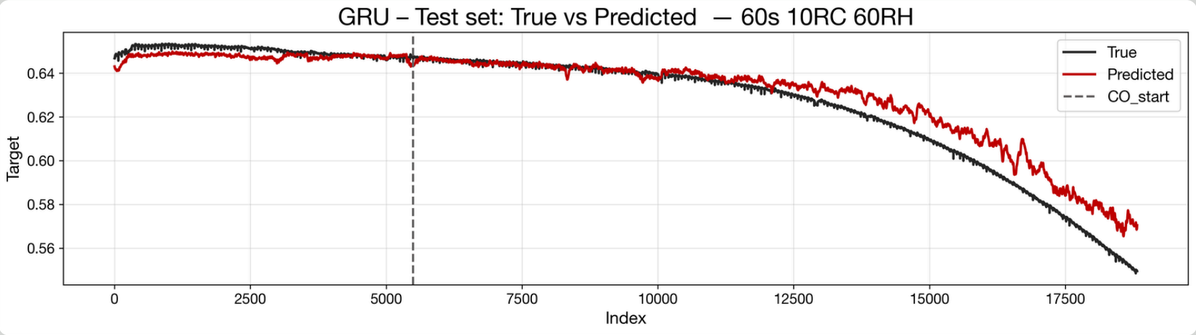


Leave one experiment out

Test set: 60s 10RC 60RH

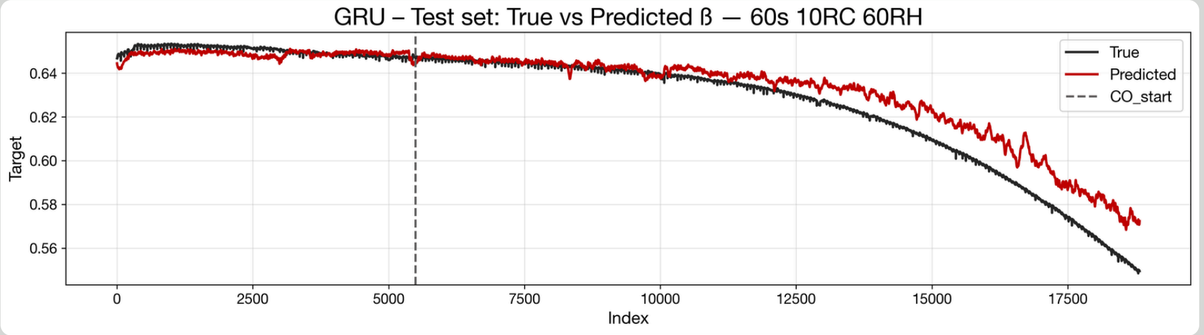
✗ CO | ✗ ß

MSE: 0.1077; MAE: 0.233; RMSE: 0.3282; $R^{2:}$ 0.9274



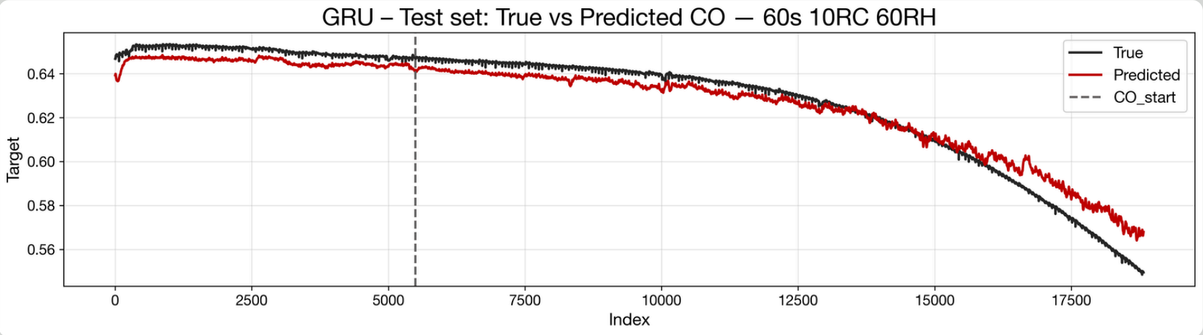
✗ CO | ✓ ß

MSE: 0.1498; MAE: 0.2722; RMSE: 0.387; $R^{2:}$ 0.8988



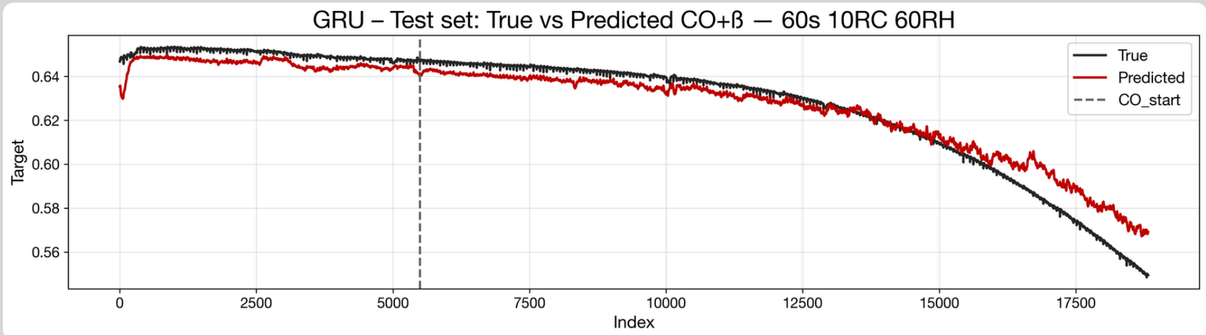
✓ CO | ✗ ß

MSE: 0.084; MAE: 0.2462; RMSE: 0.2898; $R^{2:}$ 0.9435



✓ CO | ✓ ß

MSE: 0.099; MAE: 0.2538; RMSE: 0.3146; $R^{2:}$ 0.9334

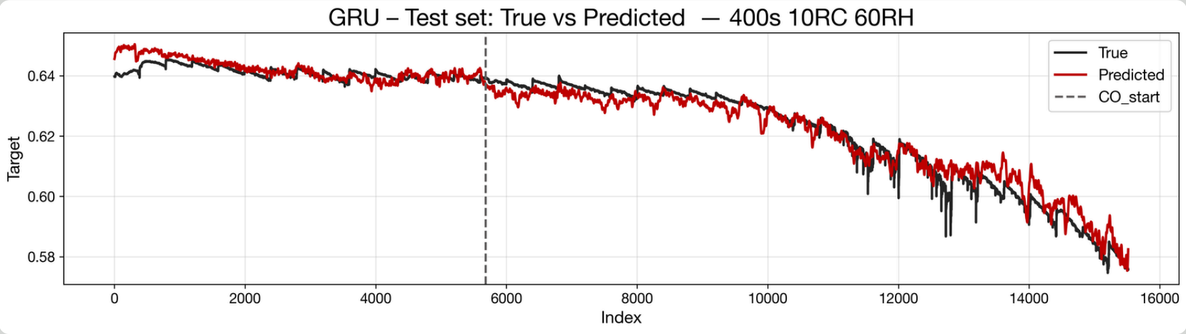


Leave one experiment out

Test set: 400s 10RC 60RH

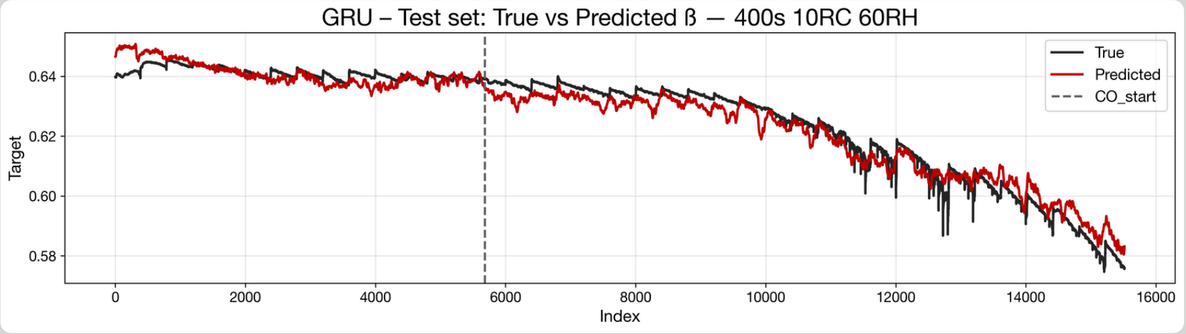
✗ CO | ✗ ß

MSE: 0.0243; MAE: 0.1186; RMSE: 0.1558; $R^{2:}$ 0.9592



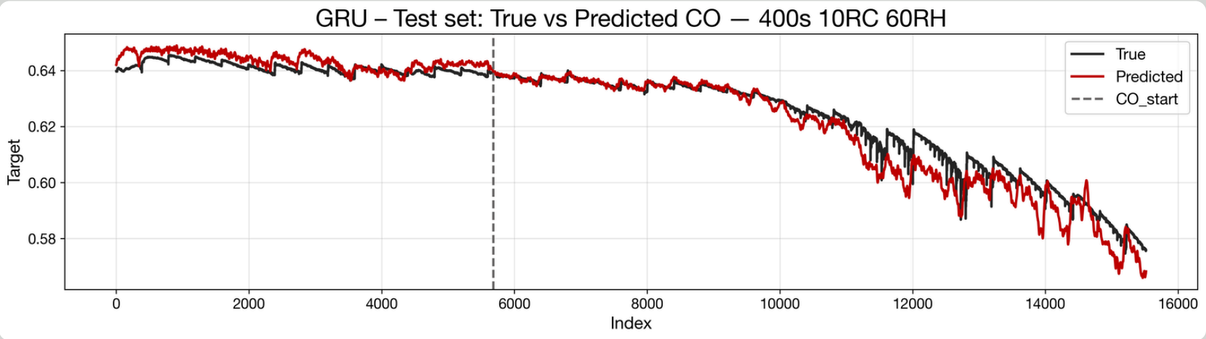
✗ CO | ✓ ß

MSE: 0.0265; MAE: 0.1328; RMSE: 0.1628; $R^{2:}$ 0.9557



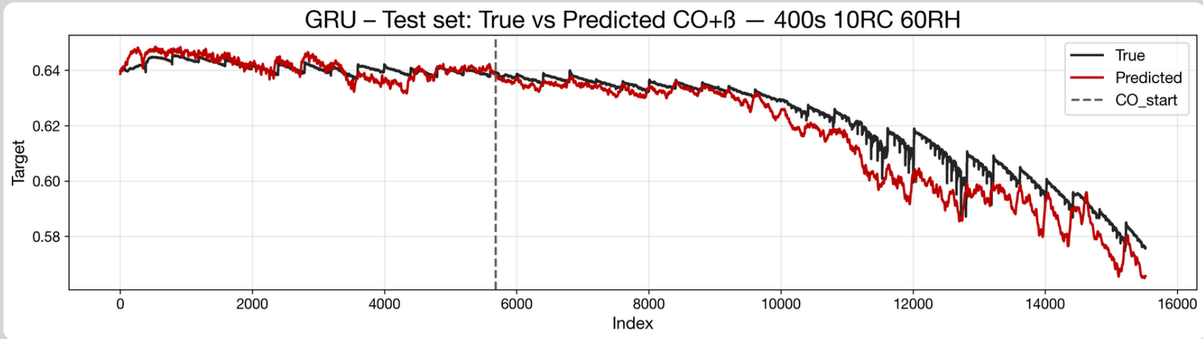
✓ CO | ✗ ß

MSE: 0.0487; MAE: 0.1604; RMSE: 0.2207; $R^{2:}$ 0.9187



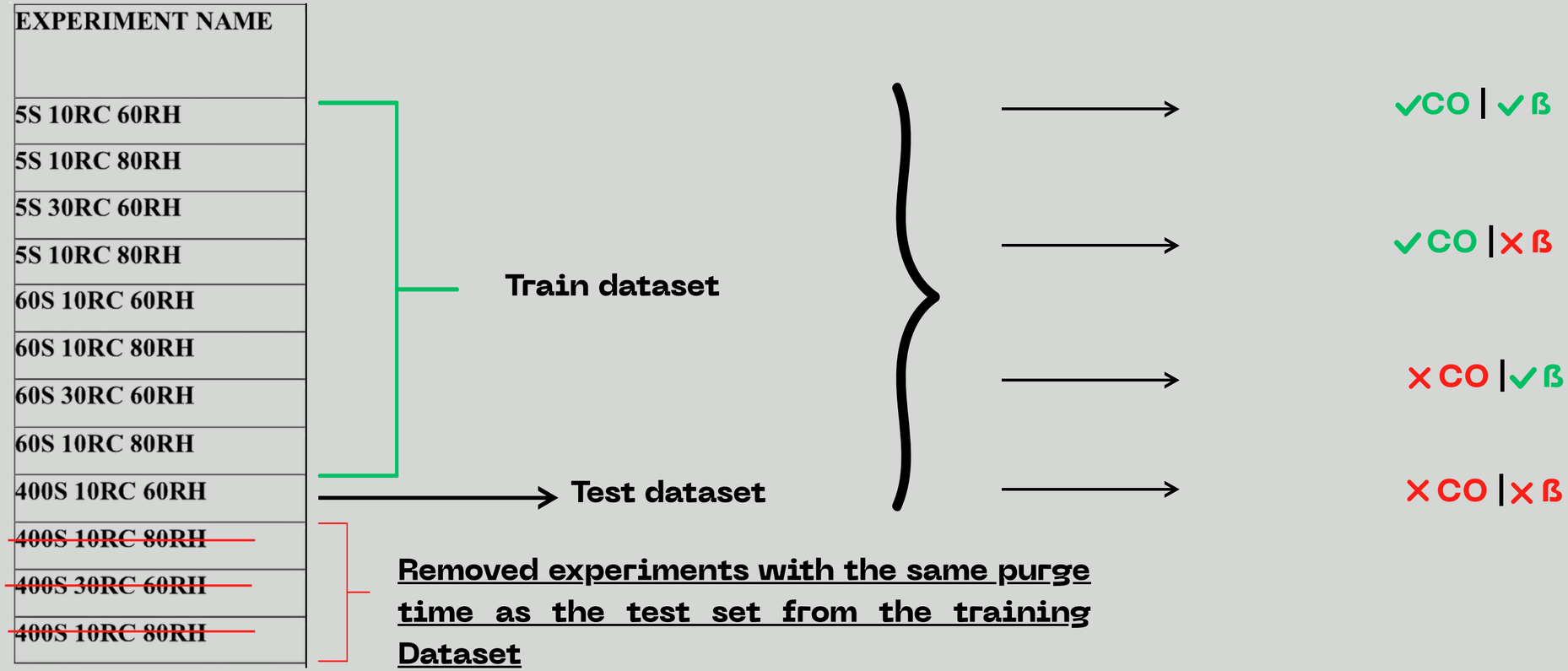
✓ CO | ✓ ß

MSE: 0.0871; MAE: 0.2157; RMSE: 0.2951; $R^{2:}$ 0.8543



Leave one purge out

We asked whether these models are able to produce meaningful predictions when confronted with operating conditions they have never seen before.



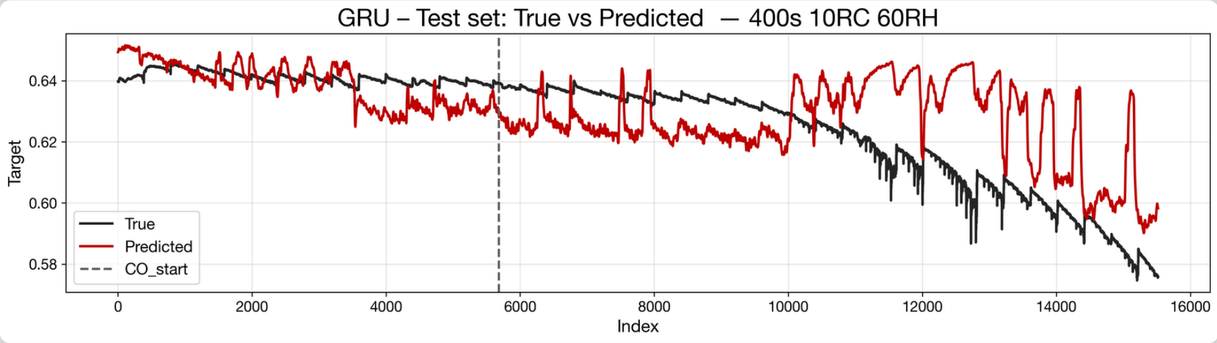
Removed experiments with the same purge time as the test set from the training Dataset

Leave one purge out

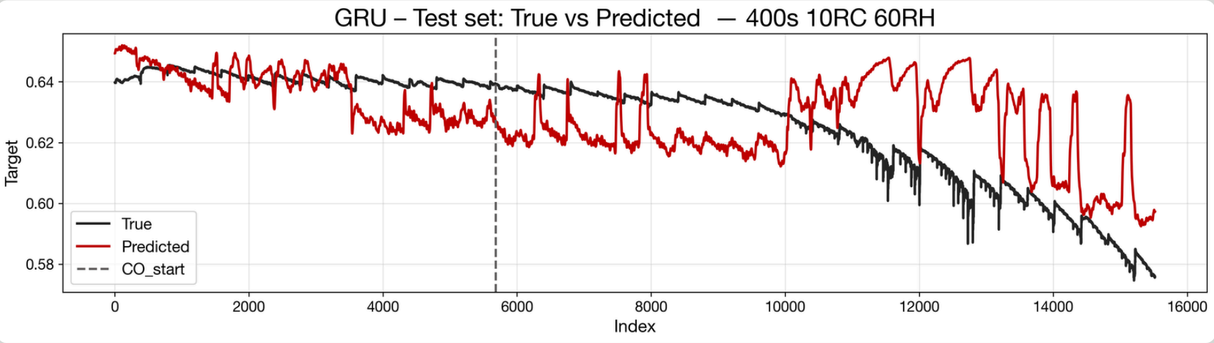
Test set: 400s 10RC 60RH

↳ **NOT in train**

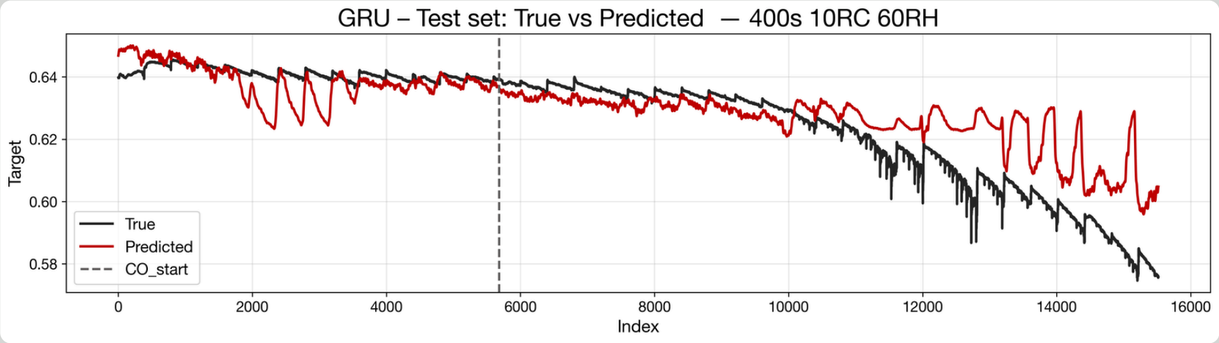
✗ CO | ✗ ß



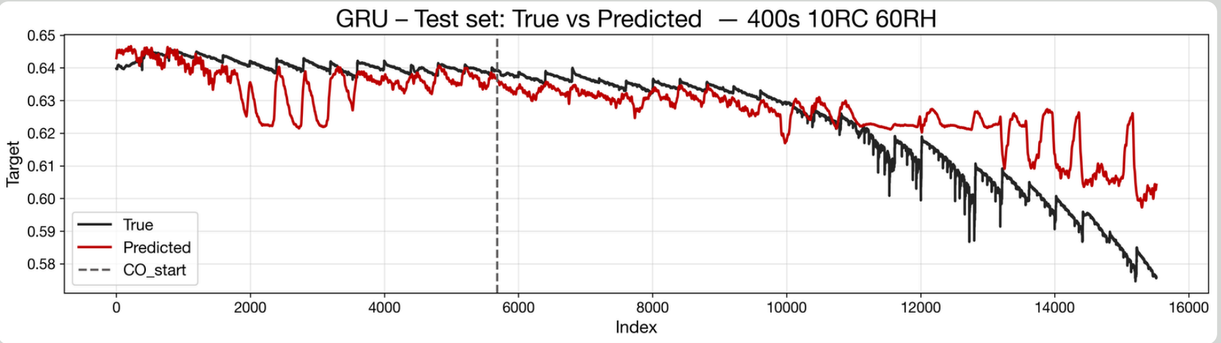
✗ CO | ✓ ß



✓ CO | ✗ ß



✓ CO | ✓ ß



Conclusions

- Deep learning models provide a viable alternative to physics-based approaches for PEMFC voltage degradation, given sufficiently diverse training conditions. heterogeneous across operating conditions.
- Their robustness is limited when facing unseen operating regimes.

Based on the results, we observed a clear drop in performance under unseen conditions, indicating a distribution shift.
To address this, we applied a domain adaptation method called CORAL, which aligns the statistical structure of the data across domains.

Domain Adaptation

Data Loaders

- **Source set (labeled):** used for regression + domain alignment (only healthy region)
- **Target set (unlabeled)**: used only for domain adaptation (same healthy region)
- **test set**: full experiment. never seen during training

→ Alignment is performed only on the stable “healthy” phase

Scaling strategy:

- Source:
  - regression: inputs are standardized using statistics from the full source domain, while the target is normalized using a scaler derived from the source data
  - domain adaptation: Alignment is performed on the healthy subset
- Target: only inputs are scaled from the healthy subset.
- Test: inputs are scaled using combined the whole source domain and the target healthy subset, while the outputs use the same source scalers

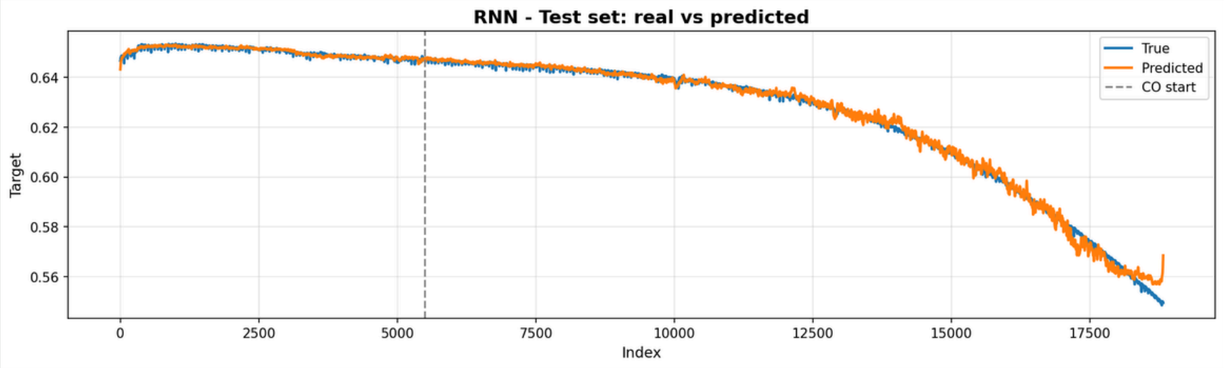
$$\mathcal{L}_{total} = \mathcal{L}_{reg}(X_s, Y_s) + \lambda * \mathcal{L}_{DA}(X_s^{healthy}, X_t^{healty})$$

Results

source: 60s 10RC 60RH
target/test: 400s 10RC 60RH

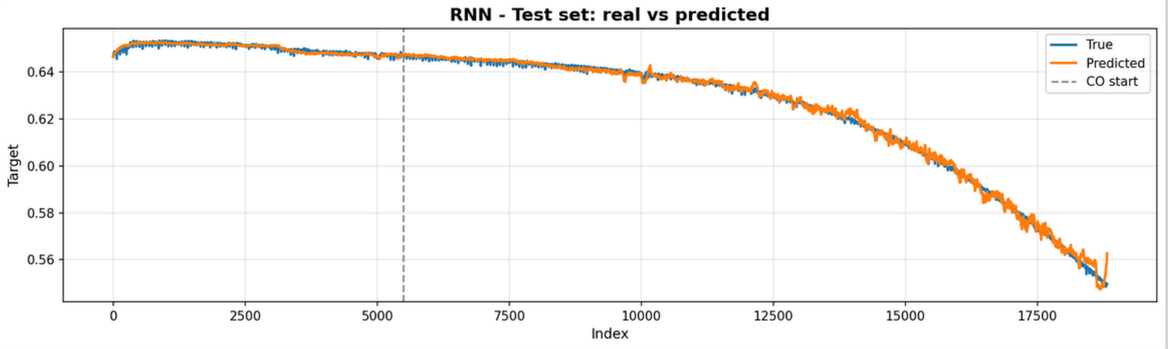
Source

$\lambda = 0$

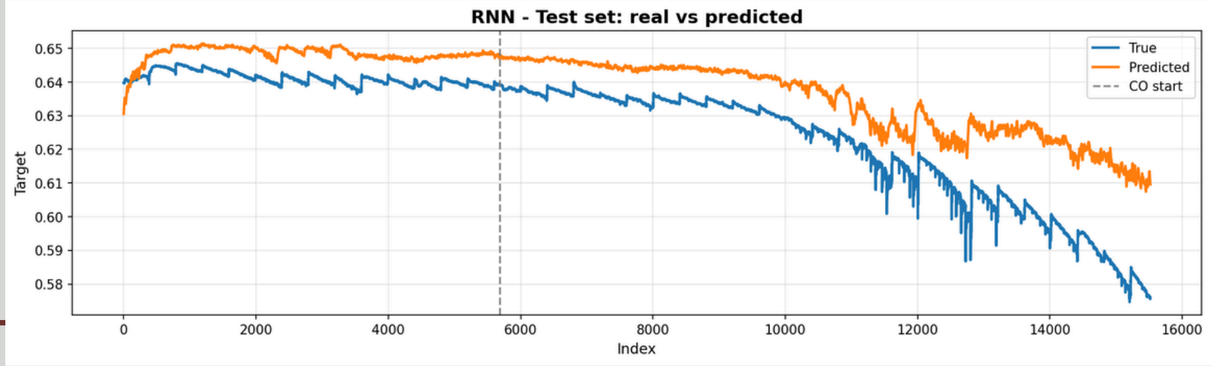


Source

$\lambda = 40$



Target



Target

